Indian Institute of Technology, Bhilai



CSP203: Software Tools & Technologies Laboratory

Lab Activity - 14

Full Marks: 30 *Deadline: 5:30 PM, 16-10-2025*

Submission Instructions:

1) Answer all the questions.
2) You should prepare a zip file which will contain all the .py, and .pdf files of the task in their respective folders.
3) Each python script file should be placed in its respective task folder (e.g., Task1, Task2..).
4) Your zip file should be named as <Roll No>_<Name>_<Lab Activity No>.zip
5) Example: B24CS055_Shareen_14.zip
6) **Plagiarism is strictly prohibited.**

Objectives:

1) Evaluate and compute complex mathematical functions accurately using Python.
2) Visualize spatial temperature data through smooth 2D heatmaps using Matplotlib.
3) Customize plots with clear labels, titles, legends, and professional font styles.
4) Create multi-plot layouts using subplots for comparative visualization of functions.
5) Integrate computation and visualization to interpret mathematical and spatial data effectively.

Task1: [10 Marks]

1)You are given 10 complex mathematical functions defined as follows:

1. $f_1(x) = sin(x) \cdot e^{-0.1x} + \frac{cos(x^2)}{x^2+1}$
2. $f_2(x) = ln(x^2 + 1) + \sqrt{sin(x) + 2}$
3. $f_3(x) = e^{-sin(x)} \cdot cosh(0.5x) - tan^{-1}(x)$
4. $f_4(x) = \frac{x^3-2x+1}{x^2+1} + sin(3x) \cdot cos(x^2)$
5. $f_5(x) = \sqrt{|x|} \cdot sin(\frac{1}{x+0.1}) + ln(x^2 + 2)$
6. $f_6(x) = \frac{sin(x)\cdot cos(3x)}{1+e^{-x}} + \sqrt[3]{x^2 + 1}$
7. $f_7(x) = e^{-x^2} \cdot sin(5x) + \frac{ln(x+2)}{x+1}$
8. $f_8(x) = tanh(x) + \frac{cos(x)}{1+x^2} + e^{-\frac{x}{5}}$
9. $f_9(x) = sin(x^3) \cdot e^{-0.5x} + \frac{1}{1+x^4}$
10. $f_{10}(x) = \sqrt{|sin(x)|} + cosh(\frac{x}{2}) - \frac{ln(x^2+1)}{x+1}$

Your task is to:

1. Write a Python code that evaluates all these functions for each value of *x* from the list: *x*={0.5, 1.0, 1.5, 2.0, 2.5}
2. Print the function values for each *x* clearly in the terminal in a tabulated or readable format.

Hints:

1. Define all functions explicitly in Python.
2. Use a loop or repeat commands in Python for the 5 different *x* values.

3. No plotting is required.

Output Requirement:

The .py file with name ‘numpy_functions.py’

Task 2: [10 Marks]

Problem Statement:

You are given a set of temperature readings collected from various (*x*,*y*) coordinates within a rectangular room. These readings represent the spatial distribution of temperature across the room. Your task is to visualize this temperature data as a smooth 2D heatmap using Matplotlib in Python to better understand how temperature varies across different parts of the room.

Temperature Data:

Use the following temperature readings directly in your Python code as a list of tuples:

```
temperature_points = [
    (0, 0, 21.5), (1, 0, 22.0), (2, 0, 22.8), (3, 0, 23.5), (4, 0, 24.0),
    (5, 0, 23.2), (6, 0, 22.5), (7, 0, 21.9), (8, 0, 21.2),
    (0, 1, 21.6), (1, 1, 22.1), (2, 1, 23.0), (3, 1, 24.1), (4, 1, 24.6),
    (5, 1, 23.9), (6, 1, 22.8), (7, 1, 22.0), (8, 1, 21.3),
    (0, 2, 21.7), (1, 2, 22.3), (2, 2, 23.2), (3, 2, 24.5), (4, 2, 24.9),
    (5, 2, 24.1), (6, 2, 23.0), (7, 2, 22.2), (8, 2, 21.5)
]
```

Requirements:

1. Write a Python script to generate a realistic and smooth heatmap of the temperature distribution in the room using Matplotlib.
2. Your script must:
    a. Use the provided temperature data from the list above.
    b. Perform interpolation to create a smooth temperature surface.

c. Plot the heatmap using a color palette transitioning from blue (cold) to red (hot).
d. Include axis labels ("X Position (m)", "Y Position (m)").
e. Add a colorbar labeled "Temperature (°C)".
f. Add a descriptive plot title "Room Temperature Heatmap".
g. Use Times New Roman font for all text.
h. Save the output figure as temperature_heatmap.png.

Expected Output:

1. A PNG image showing a temperature heatmap where:
2. Colors represent temperature intensity, ranging smoothly from cold (blue) to hot (red).
3. Axes reflect the physical coordinates of the room.
4. Smooth gradients reveal gradual or sharp temperature variations.

Output Requirement:

1. Figure saved as a .png file with the name 'room_temperature.png'.

Task 3: [10 Marks]

Problem Context:

As a part of your data visualization practice, you are tasked with analyzing and visualizing the behavior of different mathematical functions over a fixed interval. Instead of plotting each function in separate figures, you will present all plots in a single figure using Matplotlib's subplot feature.

Objective:

Create a 2×2 grid of subplots to visualize the following four mathematical functions:

1. $f_1(x) = sin(x)$
2. $f_2(x) = cos(x)$
3. $f_3(x) = e^{-0.1x^2} \cdot sin(2x)$

4. $f_4(x) = x \cdot sin(x)$

Requirements:

1. Use NumPy to create an array x of 400 evenly spaced values in the range from -10 to 10.
2. Define each of the above functions in Python.
3. Use Matplotlib's subplots() to create a 2×2 grid layout.
4. Plot each function in a separate subplot using different colors.
5. Each subplot must have:
    a. A title indicating the function name.
    b. Gridlines for better readability.
    c. Add a main title to the figure: "Subplot of Four Mathematical Functions"
6. Ensure plots do not overlap using tight_layout().
7. Display the final plot using plt.show().

Expected Output:

A single figure containing four neatly arranged subplots, each showing the curve of one function over the given x-range. Each plot should be clearly labeled with its corresponding function name, and the layout should be aesthetically clean and readable. The figure should be saved as .pdf file with the name 'multiplot.pdf'.